

Dr. Sandeep Krishnan

Theoretical Physicist — PhD, TIFR Mumbai

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Profile

Theoretical physicist (PhD, TIFR Mumbai 2025; postdoc at ICTS Bengaluru) specializing in numerical methods for quantum field theory: large-scale Monte Carlo simulation, stochastic differential equations, high-performance C++/Python. Seeking to apply these methods in quantitative finance.

Skills

Mathematics: Stochastic processes, SDEs, measure-theoretic probability, statistical inference, numerical PDEs, Markov Chain Monte Carlo

Code: C++17 (HPC, MPI/OpenMP), Python (NumPy/SciPy/JAX), Mathematica, Git, Linux clusters

Finance: Self-study: Hull, Shreve (Stochastic Calculus for Finance I–II), QuantLib exercises

Experience

Postdoctoral Fellow — ICTS-TIFR, Bengaluru

Oct 2025 – present

- Lead developer of a lattice simulation code (C++/MPI) running on 10k+ cores at national HPC facilities; 40% runtime reduction via cache-aware data layout.
- Developed variance-reduction techniques for MCMC estimators (importance sampling, control variates).

PhD Researcher, Theoretical Physics — Tata Institute of Fundamental Research, Mumbai

Aug 2020 – Sep 2025

- Thesis on non-perturbative methods in gauge theories; 5 publications, 2 first-author in Physical Review D.
- Supervised visiting students on stochastic simulation methods.

Education

PhD Theoretical Physics — TIFR, Mumbai, 2020 – 2025

BS-MS (Physics), Gold Medal — IISER Pune, 2015 – 2020

Selected publications

Two first-author papers in Physical Review D (2023, 2024); full list on request.

Languages known

Malayalam (native), English (fluent), Hindi (fluent), Tamil (working)